

Week 1 Data Gathering, Cleaning & Preprocessing

Junior Data Science Internship
Online Retail Dataset

1. Objective

The objective of Week 1 was to collect and inspect a publicly available retail dataset, identify data-quality issues, perform cleaning and preprocessing, validate the results, and prepare a cleaned CSV dataset for further analysis.

2. Dataset Overview

The project uses an Online Retail transaction dataset stored as **Online Retail.xlsx**. The dataset contains 8 columns: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, and Country.

3. Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, and Visual Studio Code were used. The final output was exported as CSV.

4. Data Cleaning Methodology

Issue	Treatment
Exact duplicates	Removed using drop_duplicates()
Missing Description	Replaced with 'Unknown Product'
Missing CustomerID	Retained as missing; no artificial ID created
InvoiceDate	Converted to datetime
Quantity	Converted to numeric; negative values retained
UnitPrice	Converted to numeric; values ≤ 0 removed
Outliers	Investigated using IQR; not automatically removed
Normalization	Not applied; business values kept interpretable

5. Duplicate Handling

The original dataset contained 5,263 exact duplicate rows. These were removed using Pandas drop_duplicates(). The final dataset was reloaded after export and confirmed to contain 0 exact duplicate rows.

6. Invalid UnitPrice Handling

The investigation identified 2,515 records with UnitPrice equal to zero and 2 records with a negative UnitPrice. The two negative-price records were associated with 'Adjust bad debt' entries rather than normal product sales. All records with UnitPrice ≤ 0 were removed.

7. Missing Values

Missing product descriptions were represented as 'Unknown Product' to preserve otherwise usable transactions. Missing CustomerID values were retained because customer identifiers cannot be reliably inferred from the available data.

8. Negative Quantity

Negative Quantity values were retained because they may represent returns, cancellations, or reversal transactions. They were not treated as automatically invalid.

9. Outlier Analysis

The Interquartile Range (IQR) method was used to identify statistical outliers in Quantity and UnitPrice. Outliers were investigated but not automatically removed because unusual retail transactions can be legitimate bulk or high-value transactions.

10. Before and After Summary

Metric	Result
Original rows	541,909
Columns	8
Exact duplicate rows removed	5,263
Zero UnitPrice records identified	2,515
Negative UnitPrice records identified	2
Total UnitPrice <= 0 removed	2,517
Final cleaned rows	534,129
Final duplicate rows	0
Final invalid UnitPrice rows	0

11. Final Validation

The cleaned CSV was reloaded and validated. The final dataset contained 0 exact duplicate rows and 0 records with UnitPrice <= 0. InvoiceDate was converted to datetime. CustomerID missing values were intentionally retained.

12. Project Deliverables

The Week 1 project includes the cleaning notebook, reusable Python cleaning script, cleaned CSV dataset, README documentation, requirements file, and this PDF report.

13. Conclusion

The raw retail dataset was systematically inspected, cleaned, validated, and exported. The workflow focused on reproducibility and evidence-based treatment of data-quality issues, providing a reliable foundation for future analysis.